Howard Math 273, HW# 2,

Fall 2026; Instructor: Sam Hopkins; Due: Friday, October 2nd

1. (a) Let $od(n)$ denote the number of partitions of n into distinct odd parts. Show that $\sum_{n\geq 0} od(n)x^n = \prod_{k\geq 1}(1+x^{2k-1})$.

   (b) Let $ee(n)$ denote the number of partitions of n with an even number of even parts, and $oe(n)$ denote the number of partitions of n with an odd number of even parts. Show that $\sum_{n\geq 0}(ee(n)-oe(n))x^n = \prod_{k\geq 1}\frac{1}{1+(-x)^k}$.

   (c) Prove that $od(n) = ee(n) - oe(n)$ for all $n \geq 0$. **Hint**: use parts (a) and (b).

2. Prove that, for any $n \geq 0$,
$$\sum_{k=0}^{n}\binom{2k}{k}\binom{2(n-k)}{n-k} = 4^n.$$

   Hint: We discussed the generating function $\sum_{n\geq 0}\binom{2n}{n}x^n$ of the central binomial coefficients. How can you use what we proved about this generating function to deduce the desired result?

3. Let $\mathrm{ODD}(n)$ denote the subset of permutations in the symmetric group S_n that have no cycles of even size. Prove that, for any $n \geq 1$,
$$\sum_{\sigma\in \mathrm{ODD}(n)} 2^{\#\mathrm{cycles}(\sigma)} = 2\cdot n!.$$

   Hint: Recall that we showed
$$\sum_{n\geq 0}\left(\sum_{\sigma\in S_n} t_1^{c_1(\sigma)}t_2^{c_2(\sigma)}\cdots t_n^{c_n(\sigma)}\right)\frac{x^n}{n!} = e^{t_1\frac{x}{1}+t_2\frac{x^2}{2}+t_3\frac{x^3}{3}+t_4\frac{x^4}{4}+\cdots},$$
   where $c_k(\sigma)$ is the number of cycles of σ of size k. How can you use this generating function identity to deduce the desired result?
   Difficult bonus (don't have to do): Give a *bijective* proof of $\sum_{\sigma\in\mathrm{ODD}(n)} 2^{\#\mathrm{cycles}(\sigma)} = 2\cdot n!$.

4. Recall that a permutation $\sigma \in S_n$ is called a *derangement* if it has no fixed points, i.e., $\sigma(i) \neq i$ for all $i = 1,\ldots,n$. Meanwhile, $\sigma \in S_n$ is called an *involution* if it is its own inverse, i.e., σ^2 is the identity permutation. Give a simple product formula for the number of permutations in the symmetric group S_n that are simultaneously derangements and involutions. Explain why your formula is correct.